

NMP2L2896
January 10, 2025

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Nine Mile Point Nuclear Station, Unit 2
Renewed Facility Operating License No. NPF-69
Docket No. 50-410

Subject: Supplement to NMP2 Licensee Event Report 2024-002, Automatic Reactor
Scram on Turbine Trip due to Failed Breaker

In accordance with the reporting requirements contained in 10 CFR 50.73(a)(2)(iv)(A), please find enclosed NMP2 Licensee Event Report (LER) 2024-002, Automatic Reactor Scram on Turbine Trip due to Failed Breaker.

There are no regulatory commitments contained in this letter.

Should you have any questions regarding the information in this submittal, please contact Matt Rossi, Site Regulatory Assurance Manager, at (315) 349-7012.

Respectfully,

Crawford, Carl Digitally signed by Crawford, Carl
Date: 2025.01.09 07:29:55 -05'00'

Carl J. Crawford
Plant Manager, Nine Mile Point Nuclear Station

CJC/MLR

Enclosure: NMP2 Licensee Event Report 2024-002, Automatic Reactor Scram on Turbine
Trip due to Failed Breaker

cc: NRC Regional Administrator, Region I
NRC Resident Inspector
NRC Project Manager

Enclosure

NMP2 Licensee Event Report 2024-002,
Automatic Reactor Scram on Turbine Trip due to Failed Breaker
Nine Mile Point Nuclear Station, Unit 2

Renewed Facility Operating License No. NPF-69



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollects.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name
Nine Mile Point Unit 2☒ 050
☐ 0522. Docket Number
4103. Page
1 OF 44. Title
Supplement to LER 2024-002-00, Automatic Reactor Scram on Turbine Trip due to failed breaker

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
09	23	2024	2024	- 002 -	01	01	09	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode

Mode 1, Power Operation

10. Power Level

100%

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ OTHER (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact

Matt Rossi, Site Regulatory Assurance Manager

Phone Number (Include area code)

(315)-349-7012

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
X	TL	BKR	G080	Yes	X	TL	BD	GE	Yes

14. Supplemental Report Expected

☒ No ☐ Yes (If yes, complete 15. Expected Submission Date)

15. Expected Submission Date

Month	Day	Year

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On 9/23/2024 failure of the Main Generator Field Circuit Breaker (41M-2EXSX01) [EXC] lead to a Generator Field Loss of Excitation [TL]. This resulted in a Main Generator Lockout which caused a Turbine Trip [TA] and automatic anticipatory RPS actuation [JC] at 07:20. High resistance across Main Generator Field Circuit Breaker (41M-2EXSX01) caused a large voltage drop across the breaker main contacts. The Auto Voltage Regulator (AVR) [90] control system responded to the lower excitation through the feedback and increased output to compensate. The increased output resulted in increased voltage drop, heat at the contacts, a subsequent fire, and Loss of Field Excitation. The resulting Turbine Trip and Turbine Stop and Control Valve closure caused an automatic anticipatory RPS actuation.

This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Nine Mile Point	☒ 050	2. DOCKET NUMBER 410	3. LER NUMBER		
	☐ 052		YEAR 2024	SEQUENTIAL NUMBER - 002	REV NO. - 00

NARRATIVE**I. DESCRIPTION OF EVENT:****A. PRE-EVENT PLANT CONDITIONS:**

Prior to the event, NMP2 was in Mode 1 operating at 100% power.

B. EVENT:

On 9/23/2024 failure of the Main Generator Field Circuit Breaker (41M-2EXSX01) [EXC] lead to a Generator Field Loss of Excitation [TL]. This resulted in a Main Generator Lockout which caused a Turbine Trip [TA] and automatic anticipatory RPS actuation [JC] at 07:20. High resistance across Main Generator Field Circuit Breaker (41M-2EXSX01) caused a large voltage drop across the breaker main contacts. The Auto Voltage Regulator (AVR) [90] control system responded to the lower excitation through the feedback and increased output to compensate. The increased output resulted in increased voltage drop, heat at the contacts, a subsequent fire, and Loss of Field Excitation. The resulting Turbine Trip and Turbine Stop and Control Valve closure caused an automatic anticipatory RPS actuation. This event is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

C. INOPERABLE STRUCTURES, COMPONENTS, OR SYSTEMS THAT CONTRIBUTED TO THE EVENT:

None.

D. DATES AND APPROXIMATE TIMES OF MAJOR OCCURRENCES:

September 23, 2024:

07:20: All Condensate Booster pumps, the 'A' Feedwater pump, and the 'C' Feedwater (FWS) Pump, tripped. In addition, the A and B recirc pumps tripped from fast speed. This is all indicating that a slow transfer occurred. It was verified that generator output breaker R925 and R230 auto opened as expected due to the generator lockout relay 86-1-2SPGZ01 actuating (Reference EE-001B and N2-OP-68). Reactor water level decreased below Level 2 (L2, 108.8") reaching 98.5" due to the trip of the 'A' and 'C' Feedwater pumps. This resulted in multiple Primary Containment Isolation System (PCIS) isolations. Groups 2, 3, 6, 7, 8, and 9 isolated as expected per N2-OP-83. The L2 actuations also caused Reactor Core Isolation Cooling (RCIC), High Pressure Core Spray (HPCS) and Division 3 diesel generator to auto start, as designed. RCIC and HPCS injected to restore level above L2. The Reactor Building (RB) ventilation system isolated and Standby Gas Treatment (SBGT) and Control Room Emergency Filtration (CREF) systems started as designed. These are all expected responses for reaching a L2. Reactor pressure reached a peak of 1102 psig. Reactor pressure was controlled by the turbine bypass valves. (Reference IR 04804263)

07:20: HPCS and RCIC were used to control reactor water level between 160" and 200".

07:38: Operations received report from RP tech that they see and smell smoke on the Turbine Building Elevation 306'. Operator was dispatched to investigate.

07:39: Operations reported visible flames on the Main Generator Field Circuit Breaker. Operator attempted to extinguish the fire using portable CO2 extinguishers.

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1. FACILITY NAME

Nine Mile Point

**050****052****2. DOCKET NUMBER**

410

3. LER NUMBER**YEAR**

2024

SEQUENTIAL**NUMBER**

002

REV**NO.**

00

NARRATIVE

07:55: HPCS was secured and RCIC alone was used to control reactor water level until the station was within the capacity of a Condensate booster pump.

08:03: Operations reported fire was extinguished and no EAL thresholds were met.

08:05: Breakers 2NJS-US3-6B and 2NJS-US3-13B were opened to remove power from the AVR to the exciter.

09:19: Electrical Maintenance (EMD) identified the 'B' station battery had a +145 VDC ground. OPS identified that the AVR and exciter had control power from the 'B' station battery.

10:02: Breaker 2BYS-PNLB107-14, the B station battery feed to the AVR, was opened per N2-OP-68B step G.1.7. Prior to opening the breaker, 'B' battery indicated +145 VDC to ground. After opening 2BYS-PNLB107-14, 'B' battery remained +145 VDC to ground.

10:55: The trip and close fuses for the exciter breaker were removed. This resulted in the 'B' station battery ground voltages returning to normal values.

1445: RCIC was secured and Reactor level control was transitioned to 'B' condensate booster pump.

E. METHOD OF DISCOVERY:

This event was self-revealed when the Turbine Control Valve Fast Closure signals actuated RPS 'A' and 'B', resulting in unit scram.

F. SAFETY SYSTEM RESPONSES:

All safety systems responded per design.

II. CAUSE OF EVENT:

The cause of the event was a mechanical fault internal to the breaker caused a high resistance connection to develop ultimately resulting in loss of the generator field.

The internals of the breaker melted during the event destroying evidence. The internal damage prevented the identification of a specific sub-component failure including potential process, or programmatic causes associated with the sub-component maintenance, construction, or testing.

III. ANALYSIS OF THE EVENT:

The scram did not have any impact to the health and safety of the public. All safety systems responded per design.

NMP2 tripped due to increased resistance in the contact of the 'A' (-) phase of the 41M field breaker. The increased resistance caused the breaker to burn further degrading the contacts, which increased the resistance until it dropped the generator excitation. This was identified by generator output voltage drop. The feedback loop from the AVR increased current to recover the generator field which caused the output voltage to increase and generator output voltage to recover.

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	☐ 052		<table border="1"><tr><td>YEAR</td><td>SEQUENTIAL NUMBER</td><td>REV NO.</td></tr><tr><td>2024</td><td>002</td><td>00</td></tr></table>	YEAR	SEQUENTIAL NUMBER	REV NO.	2024
YEAR	SEQUENTIAL NUMBER	REV NO.					
2024	002	00					

NARRATIVE

The increased current further increased temperature which also increased resistance on the 'A' (-) phase breaker contacts. The feedback loop between resistance increase, generator field voltage drop, AVR current control, and breaker contact temperature, continually increased temperature and degraded the breaker contacts. The AVR continued to respond until it hit a max current limit at which point it could not improve generator field voltage although field voltage continued to degrade. This resulted in the loss of the generator field. This tripped the 40-1-2SPGZ02 relay based on settings calculated in EC-167 which takes into account generator reactive capability. This in turn initiated the 86-1-2SPGZ01 relay which triggered the generator lockout sequence and resulted in the turbine trip and SCRAM.

IV. CORRECTIVE ACTIONS:**A. ACTION TAKEN TO RETURN AFFECTED SYSTEMS TO PRE-EVENT NORMAL STATUS:**

During the N2F47 forced outage the station replaced the 41M-2EXSX01 Main Generator Field Breaker and cubicle (panel) assembly (W/O C93993280).

B. ACTION TAKEN OR PLANNED TO PREVENT RECURRENCE:

Preventive Maintenance program changes for the breaker have been implemented with thermography performed every two weeks and monitoring of generator field voltage and current.

The Corrective Action to Prevent Recurrence is to modify the generator field circuitry to remove the generator field breaker. This will remove the component and cubicle and prevent a similar failure mechanism.

V. ADDITIONAL INFORMATION**A. FAILED COMPONENTS:**

Main Generator Field Circuit Breaker (41M-2EXSX01)

B. PREVIOUS LERs ON SIMILAR EVENTS:

None